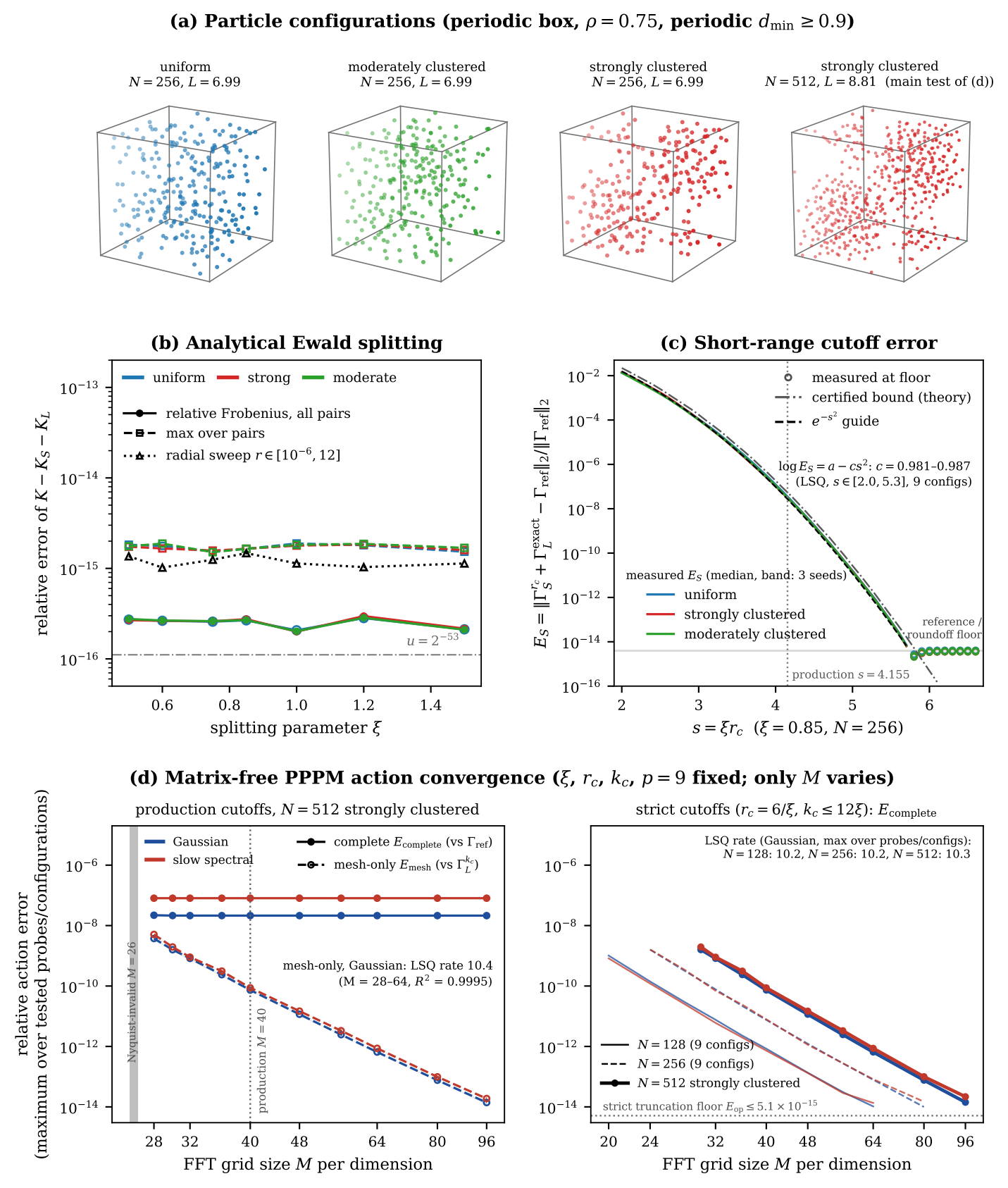

(a) Particle configurations (periodic box, $\rho = 0.75$, periodic $d_{\min} \geq 0.9$)
uniform
$N = 256$, $L = 6.99$
moderately clustered
$N = 256$, $L = 6.99$
strongly clustered
$N = 256$, $L = 6.99$
strongly clustered
$N = 512$, $L = 8.81$ (main test of (d))
(b) Analytical Ewald splitting
uniform
strong
moderate
relative Frobenius, all pairs
max over pairs
radial sweep $r \in [10^{-6}, 12]$
$u = 2^{-53}$
relative error of $K - K_S - K_L$
splitting parameter ξ
(c) Short-range cutoff error
measured at floor
certified bound (theory)
e^{-s^2} guide
$\log E_S = a - cs^2$: $c = 0.981$–0.987
(LSQ, $s \in [2.0, 5.3]$, 9 configs)
measured E_S (median, band: 3 seeds)
uniform
strongly clustered
moderately clustered
reference / roundoff floor
production $s = 4.155$
$E_S = \|\Gamma_S^{r_c} + \Gamma_L^{\text{exact}} - \Gamma_{\text{ref}}\|_2 / \|\Gamma_{\text{ref}}\|_2$
$s = \xi r_c$ ($\xi = 0.85$, $N = 256$)
(d) Matrix-free PPPM action convergence (ξ, r_c, k_c, $p = 9$ fixed; only M varies)
production cutoffs, $N = 512$ strongly clustered
Gaussian
slow spectral
complete E_{complete} (vs Γ_{ref})
mesh-only E_{mesh} (vs $\Gamma_L^{k_c}$)
mesh-only, Gaussian: LSQ rate 10.4
(M = 28–64, $R^2 = 0.9995$)
Nyquist-invalid $M = 26$
production $M = 40$
relative action error
(maximum over tested probes/configurations)
FFT grid size M per dimension
strict cutoffs ($r_c = 6/\xi$, $k_c \leq 12\xi$): E_{complete}
LSQ rate (Gaussian, max over probes/configs):
$N = 128$: 10.2, $N = 256$: 10.2, $N = 512$: 10.3
$N = 128$ (9 configs)
$N = 256$ (9 configs)
$N = 512$ strongly clustered
strict truncation floor $E_{\text{op}} \leq 5.1 \times 10^{-15}$
FFT grid size M per dimension